



SP-201 RED

CS4850 | Section 2 | Spring 2026

Multi-Agent Research & Fact-Checking Pipeline

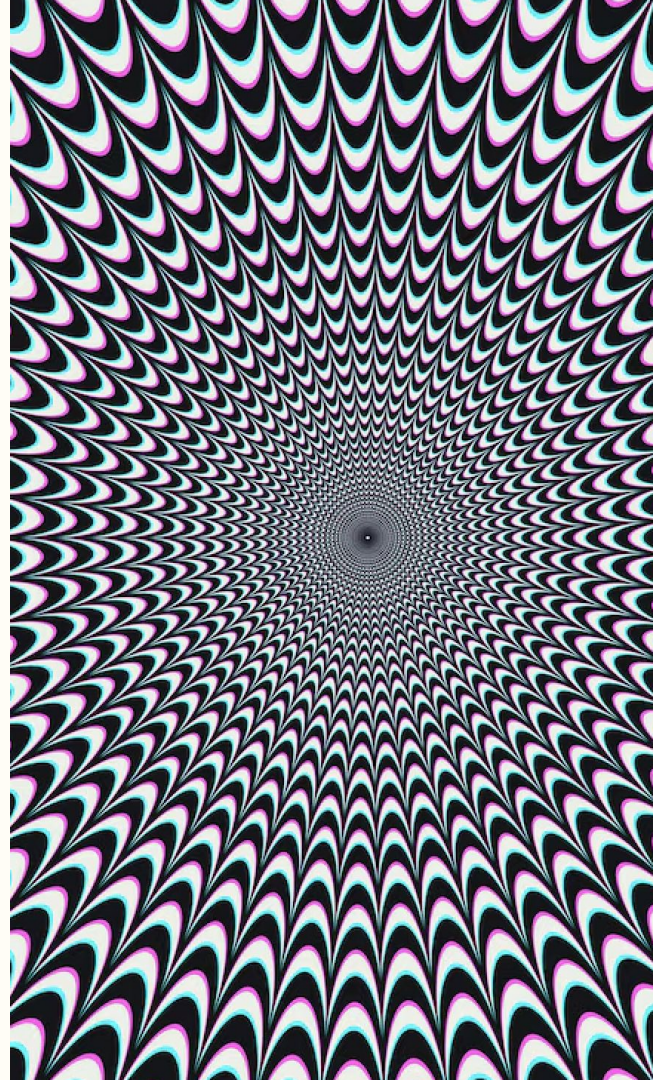
RAAID KHAN, NHA PHAN, AFRA KURUDIREK, MELIKE OZCELIK

01. The Problem

CURRENT AI SYSTEMS CAN GENERATE FLUENT AND DETAILED RESPONSES, BUT THEY OFTEN PRODUCE UNVERIFIED OR INCORRECT INFORMATION DUE TO A LACK OF RELIABLE FACT-CHECKING.

THIS MAKES THEM DIFFICULT TO TRUST FOR RESEARCH AND ACADEMIC USE, ESPECIALLY WHEN ACCURATE SOURCES AND VALIDATION ARE REQUIRED.

THEY OFTEN PRODUCE HALLUCINATIONS.



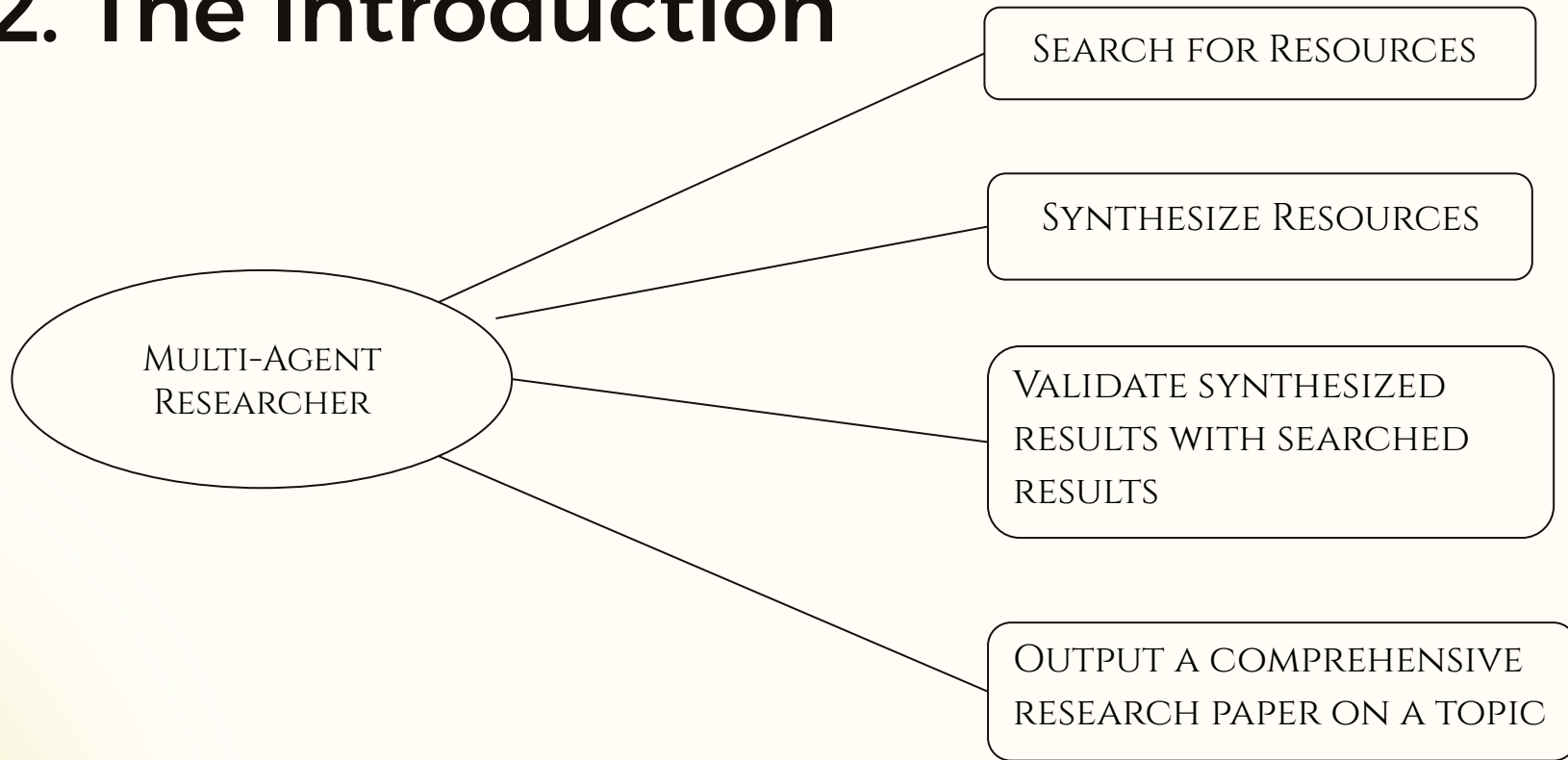
IN 2023, ~41% OF U.S. ADULTS HAD USED A GENERATIVE AI TOOL.

Year	% People Using AI Tools	% People Using AI for Fact-Checking
2023	41%	9%
2024	56%	15%
2025	67%	22%

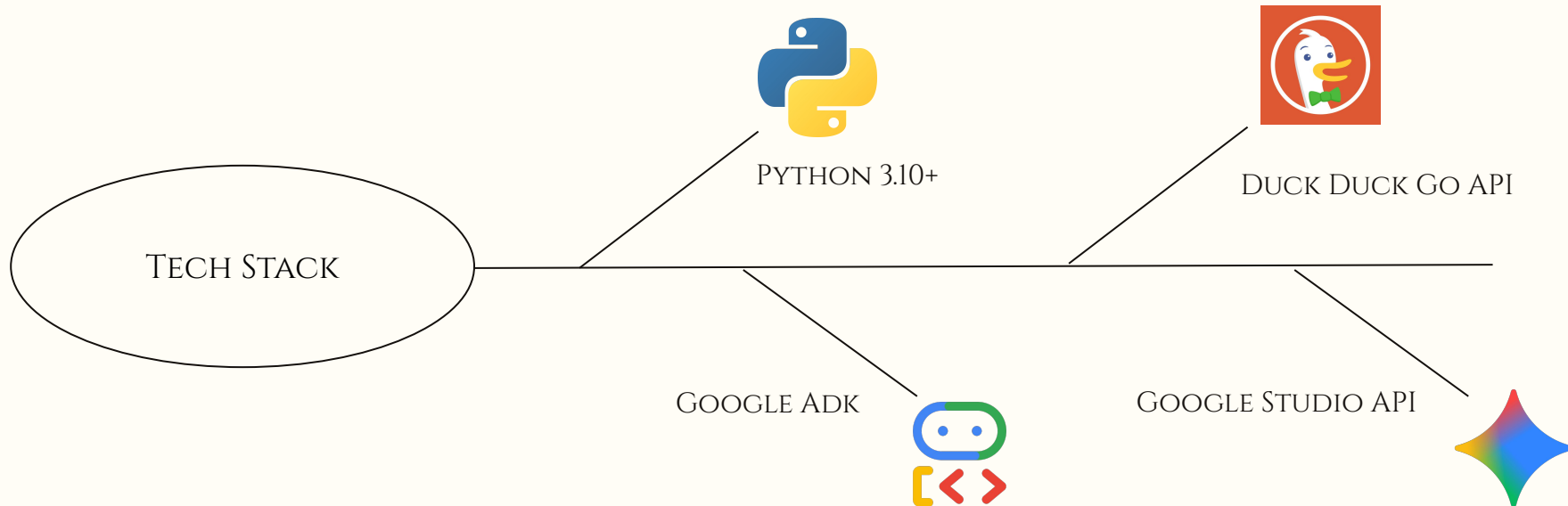
AI USAGE PERCENTAGES ARE ADAPTED FROM PEW RESEARCH CENTER DATA
ON GENERATIVE AI ADOPTION



02. The Introduction



03. The Tech Stack



04. The Breakdown

MANAGER AGENT

TAKES THE USER'S PROMPT

BREAKS IT INTO SMALLER
QUESTIONS

ASSIGNS TASKS TO OTHER
AGENTS

MANAGES THE WORKFLOW,
INCLUDING RE-RUNNING
STEPS IF ERRORS OCCUR.

SEARCH AGENT

GATHERS INFORMATION FROM
EXTERNAL SOURCES

RETRIEVES RELEVANT DATA
FROM APIS

RETURNS STRUCTURED RESULTS
WITH SOURCES AND CITATIONS.

04. The Breakdown

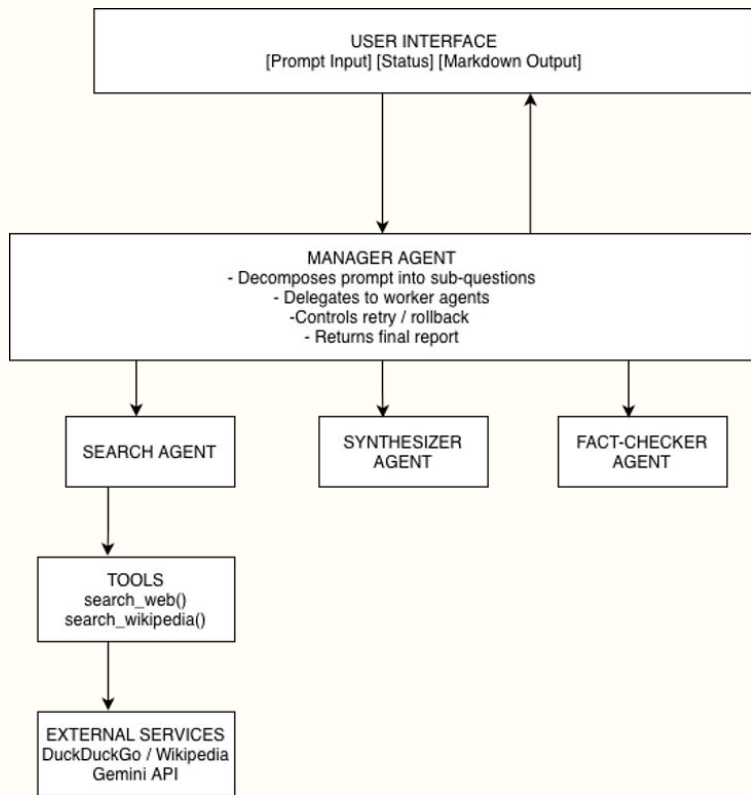
SYNTHESIZER AGENT

TAKES THAT DATA AND
TURNS IT INTO A
CLEAR, ORGANIZED
RESEARCH REPORT
WHILE KEEPING ALL
CITATION REFERENCES.

FACT-CHECKER AGENT

VERIFIES THE FINAL
OUTPUT BY COMPARING
IT TO THE ORIGINAL
SOURCES AND FLAGS ANY
UNSUPPORTED CLAIMS,
TRIGGERING
CORRECTIONS IF NEEDED.

05. The Architecture



Centralized Manager Agent controls the workflow

Decomposes user prompt into structured sub-questions

Delegates tasks to specialized agents:

- **Search Agent** → retrieves external data
- **Synthesizer Agent** → generates structured report
- **Fact-Checker Agent** → validates accuracy

Sequential pipeline ensures controlled execution

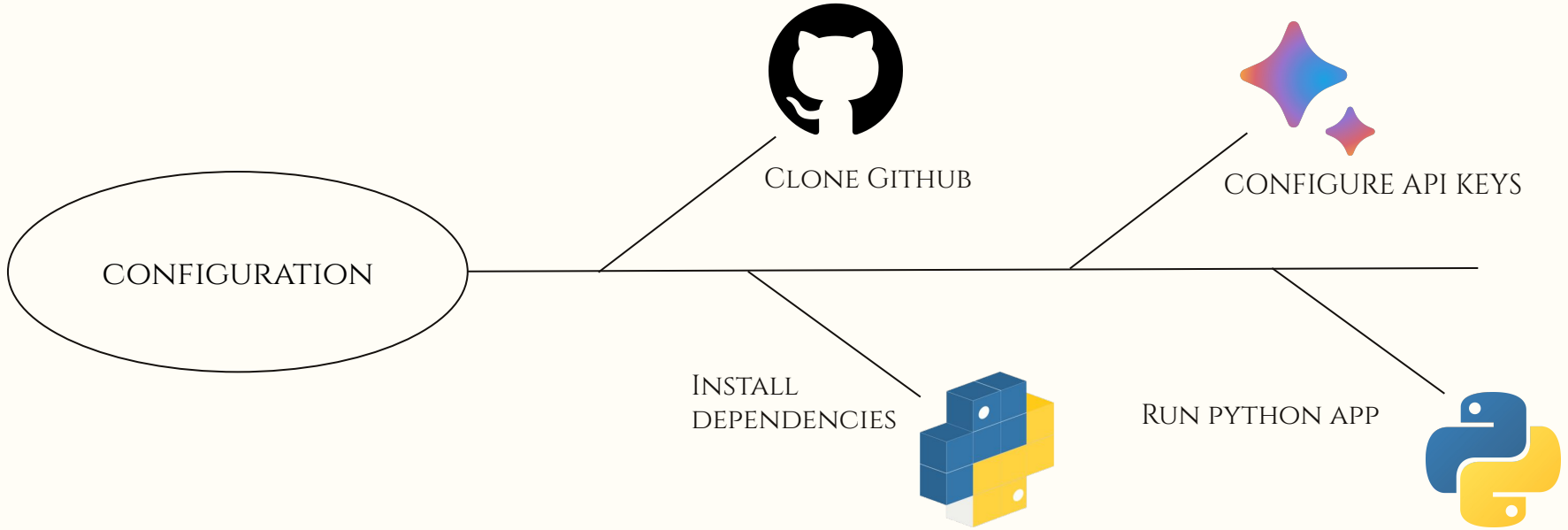
Structured data passed between agents

Rollback mechanism for error correction

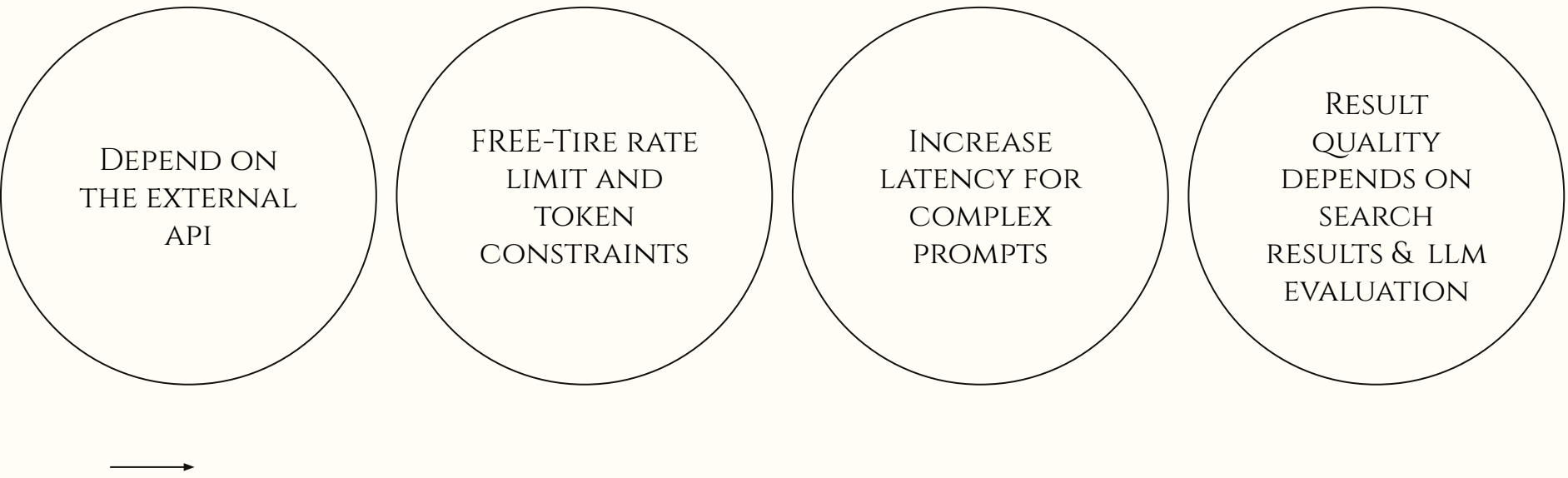
Memory

- Allows system to remember past interactions
- Stores previous queries and results
- Helps maintain context across related prompts
- Implemented using in-memory storage (Google ADK)
- Did not use cloud-based memory
- Used local in-memory storage instead

06. The Configuration



07. The Limitations



DEPEND ON
THE EXTERNAL
API

FREE-TIRE RATE
LIMIT AND
TOKEN
CONSTRAINTS

INCREASE
LATENCY FOR
COMPLEX
PROMPTS

RESULT
QUALITY
DEPENDS ON
SEARCH
RESULTS & LLM
EVALUATION

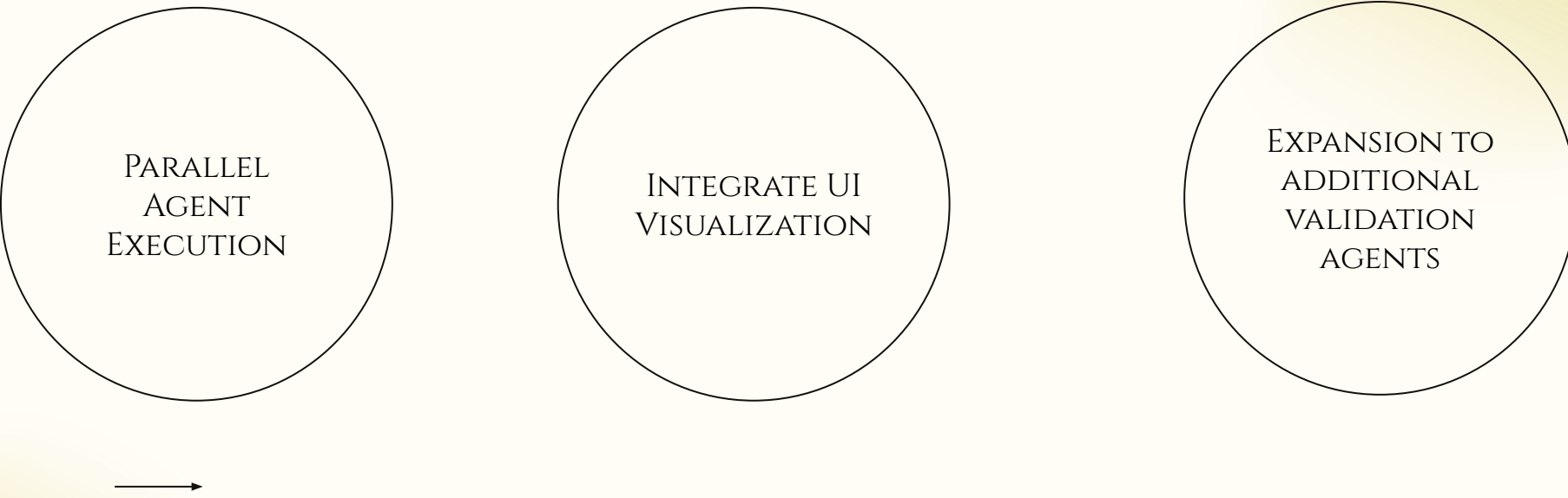
08. The Demonstration

<https://youtu.be/y9zjAtyO0Xg>

UI

<https://youtu.be/7xQYX37l3HM>

09. The Future Improvements



PARALLEL
AGENT
EXECUTION

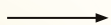
The diagram consists of three circles arranged horizontally. The first circle on the left contains the text 'PARALLEL AGENT EXECUTION'. Below it is a right-pointing arrow. The middle circle contains the text 'INTEGRATE UI VISUALIZATION'. The third circle on the right contains the text 'EXPANSION TO ADDITIONAL VALIDATION AGENTS'. The third circle is highlighted with a yellow background.

INTEGRATE UI
VISUALIZATION

EXPANSION TO
ADDITIONAL
VALIDATION
AGENTS

10. Conclusion

The current build successfully implements hierarchical multi-agent coordination, external data retrieval, structured synthesis, and automated fact-checking. The system operates within free-tier API constraints and supports a full end-to-end research workflow.



We submitted our powerpoint slides prior to this presentation.



**Thank You
for Listening!**